Jonah Wolk

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Education

Northeastern University, Boston, MA *May 2028*

Candidate for B.S. in Mechanical Engineering

GPA: 3.70

Honors: Dean's Scholarship Recipient

Classes: Heat Transfer, Fluid Mechanics, Thermodynamics, Dynamics, Statics, Physics 1-2, Calculus 1-3, Material Science

Activities: Engineers Without Borders, Music Production, ASME, EcoScholars

Work Experience

Line Design Intern | *WSP* *January 2026 - July 2026*

- Modeled, designed, and analyzed various structural designs through iterative modeling, utilizing a specialized CADD software to insure structural integrity under varying load and weather conditions while balancing material requirements, performance, and other essential constraints
- Created & evaluated finalized checkpackages with Bluebeam Revu, Excel, and industry standard engineering tools that followed a vigorous QA/QC process to ensure accuracy and compliance to client-specific design standards and codes
- Interpreted and synthesized data from various survey reports, LiDAR sources, and structural specifications in order to generate accurate computer models of transmission line routes

Projects

Automated Terrarium Climate Control System | *Personal Engineering Project* *June 2026 - Present*

- Designed and commenced construction on an intricate system to monitor and automatically regulate humidity and temperature for a bioactive terrarium utilizing Solidworks & C++
- Programmed an replicable arduino-based system to translate live sensor data, display real time conditions on a LCD screen, and trigger an pump-based automatic misting mechanism to maintain a target humidity
- Engineered a closed-loop feedback system with the goal of adjusting misting frequency based on everchanging sensor input, eliminating the need for manual monitoring

Sealed Enclosure Loudspeaker Design | *Personal Engineering Project* *October 2026 - Present*

- Modeled a sealed-box loudspeaker as an electrical circuit in LTspice using datasheet-driven Thiele-Small parameters, converting cone mass, suspension compliance, and loss into equivalent circuit components
- Ran AC analysis and parametric volume sweeps to predict impedance and system resonant frequency
- Designed an optimized 6L MDF enclosure (Qtc = 0.8) from the simulation-derived volume, with fabrication underway to compare simulated and real-world values

Compact Keychain Lock | Personal Engineering Project June 2026 - Present

- Leveraging rapid prototyping techniques to iterate a proof of concept for a lightweight, 3D-printable lock optimized for compactness and portability after identifying an underserved niche in the consumer market
- Utilizing SolidWorks and Cura to model and refine the mechanism's geometry, while balancing part tolerances and printability with functionality

Audio Amplifier Restoration | *Personal Engineering Project* *February 2026 - March 2026*

- Diagnosed and repaired a malfunctioning 1990s 130W Fender Power Chorus Amplifier purchased second hand, restoring full and complete functionality by accurately troubleshooting faulty components
- Applied knowledge of circuit analysis, soldering, and other essential techniques in order to find and replace damaged elements such as thermosisters and blown speakers

Project Lead the Way Pathway | *Co-Leader // Development Head* *August 2023 - April 2024*

- Led the design & development of a fully functional delivery robot capable of traversing complicated terrarian
- Won **Best Prototype Award** for the 2024 PLTW Regional Innovative Solutions Competition

Skills

Technical: SolidWorks, Fusion360, Soldering, AutoCAD, Projectwise, Bluebeam Revu, MATLAB, Microsoft Office, Tinkercad, C++, Arduino, Machining

Interests

Music Production, Guitar, Golf, Tennis, Cooking